SentinelAI

Industrial Safety Intelligence Platform

A compound risk intelligence platform that fuses CCTV, gas sensors, permits, maintenance, and shift data into one explainable risk engine — backed by a regulation-grounded compliance copilot and an automated emergency response orchestrator.

Project Report — Hackathon Submission

1. Executive Summary

Industrial incidents rarely stem from a single missed signal — they stem from signals that exist separately, in separate systems, and are never fused. A hot-work permit approved in one system, a gas sensor drifting toward baseline in another, a shift changeover reducing oversight in a third: individually unremarkable, together a precursor to disaster.

SentinelAI addresses this directly by building a single Compound Risk Intelligence Engine that fuses vision (CCTV/PPE), gas sensor telemetry, permit-to-work status, maintenance windows, and shift data through a multi-agent correlation pipeline, then closes the loop with a compliance copilot that answers regulatory questions strictly from ingested source documents, and an emergency response orchestrator that auto-drafts incidents and raises alerts the moment a Critical risk is detected.

Every capability described in this report is implemented and running against a real, live backend — not a static prototype. Section 6 documents exactly what was verified end-to-end.

2. Problem Statement Coverage

The platform maps directly onto the challenge's build options, consolidated where the underlying signal-fusion logic overlaps rather than duplicated:

Compound Risk Detection Engine

Multi-agent LangGraph pipeline correlating vision, gas, permit, maintenance, and shift signals via a rule-based combination matrix — explainable by design, not a black box.

Digital Permit Intelligence Agent

Shares the same correlation core; every permit is checked against live zone conditions at the moment of issuance, not only on a periodic scan.

Quality & Compliance Audit / Incident Pattern Intelligence

Extends the Compliance Copilot's RAG pipeline over ingested regulations, SOPs, and incident records.

Emergency Response Orchestrator

Matches a site-configured protocol on a Critical finding, auto-drafts a structured incident report, and raises a multi-channel alert (simulated dispatch — no real SMS/email integration in this demo).

Geospatial / Knowledge Graph extras

Scoped as roadmap items — see Section 8 — prioritized honestly behind the core fusion and response loop given the available build time.

3. Architecture

Signal sources (CCTV vision, gas sensors, permits, maintenance, shift data) feed the Compound Risk Intelligence Engine, which fans out into the Emergency Response Orchestrator and the Compliance Copilot, producing Incidents and Alerts. Multi-tenant isolation runs underneath every layer via PostgreSQL Row-Level Security, enforced independently of the ORM-level tenant filter as a defense-in-depth backstop.

API layer	FastAPI, clean api → services → ai/models architecture
Data layer	PostgreSQL (async SQLAlchemy 2 + Alembic), Row-Level Security enforced per organization
Cache / queues	Redis
Object storage	MinIO (S3-compatible) — camera evidence, uploaded compliance PDFs
Vector store	ChromaDB — the Compliance Copilot's ingested document corpus, tenant-filtered per query
AI orchestration	LangGraph multi-agent graphs — signal fusion, and RAG classify → retrieve
LLM provider	Google Gemini (chat + embeddings) via LangChain
Vision	YOLOv8 via ONNX Runtime
Frontend	React + Vite + Tailwind, plain REST over the FastAPI backend
Deployment	Docker Compose — one container per service, health-checked dependencies

4. Core Modules

4.1 Compound Risk Intelligence Engine

Four signal agents (gas sensor, permit, maintenance, shift) run in parallel inside a LangGraph StateGraph and fan into one correlation agent. The correlation logic is deliberately rule-based, not LLM-generated — a combination matrix such as "confined-space permit + gas anomaly → Critical" or "hot-work permit + gas anomaly → Critical" is deterministic and auditable, avoiding hallucinated risk reasoning in a safety-critical path. Findings feed the existing Risk Engine's time-decay and compound-aggregation pipeline as new hazard classes.

4.2 Compliance Copilot (RAG)

Uploaded PDFs are chunked, embedded via Gemini embeddings, and stored in ChromaDB with tenant isolation enforced at the query filter. A LangGraph pipeline classifies question intent and retrieves relevant chunks; generation happens only when retrieval actually found something relevant — if nothing relevant was retrieved, the system explicitly states it lacks sufficient information rather than fabricating an answer. Verified live: a question grounded in an ingested SOP returned a correctly cited, streamed answer; a question outside any ingested document was correctly refused.

4.3 Emergency Response Orchestrator

On a Critical risk snapshot, the orchestrator matches an exact hazard-type protocol, falls back to a general protocol, or explicitly reports no protocol configured — never fabricating a procedure. It then drafts a structured incident summary via Gemini, constrained strictly to the evidence at hand, with a templated fallback if the LLM call itself fails, so a safety-critical incident always gets created. A cooldown window prevents one ongoing Critical state from spawning duplicate incidents.

4.4 Multi-tenant Isolation

Every tenant-scoped table is protected at two independent layers: an ORM-level query filter (SQLAlchemy's with_loader_criteria) and PostgreSQL Row-Level Security enforced by a restricted database role that cannot bypass it. A second organization was verified to see none of the first organization's data at either layer.

5. Technology Stack

Backend	FastAPI, SQLAlchemy 2 (async), Alembic, Pydantic v2
Database	PostgreSQL 16 with Row-Level Security
AI / Orchestration	LangGraph, LangChain, Google Gemini (gemini-flash-latest, gemini-embedding-001)
Retrieval	ChromaDB
Vision	YOLOv8 (ONNX Runtime, CPU)
Frontend	React, Vite, Tailwind CSS, Axios
Infra	Docker Compose, Redis, MinIO

6. What Was Actually Verified

In the interest of technical honesty, this section states plainly what was live-tested against the running system rather than presenting invented benchmark figures. Several core modules (permit, maintenance, shift, and compound-risk correlation) are deterministic rule-based logic, not trained classifiers — they do not have a meaningful "accuracy" figure in the machine-learning sense; correctness there means their logic was exercised against constructed scenarios and produced the expected classification.

- Real Gemini generation, live: streamed token-by-token through the actual Server-Sent Events endpoint, confirmed distinct (non-templated) output on each run.
- Grounded retrieval with honest refusal: a question answerable from an ingested PDF returned a correctly cited answer; a question outside any ingested document was correctly refused rather than answered speculatively.
- Automated emergency response: a Critical risk finding produced a distinct, LLM-drafted incident summary (verified not to match the deterministic template fallback) and a corresponding alert.
- Multi-tenant isolation: verified at both the ORM filter and PostgreSQL RLS layers using two seeded organizations.
- Full Docker stack: PostgreSQL, Redis, MinIO, ChromaDB, and the API all build, migrate, and pass health checks together from a clean rebuild.

Vision inference currently runs a generic pretrained object detection model; dedicated PPE and fire/smoke detection models are architected as drop-in ONNX slots but were not trained or integrated in this build, due to the absence of a labeled dataset in the available time. This is stated here rather than implied otherwise.

7. Running the System

Backend (from the repository root):

```
cp .env.example .env   # fill in credentials, incl. GEMINI_API_KEY
docker compose up -d --build
```

Frontend (from frontend/):

```
npm install
npm run dev
```

Demo accounts: admin@sentinelai.demo and admin@acme-manufacturing.demo (a second isolated tenant), full credentials in the repository's user manual.

8. Roadmap

- Trained PPE and fire/smoke detection models to replace the current generic object detector.
- Analytics dashboard wired to real historical data (currently illustrative placeholder data).
- Lightweight knowledge graph over facility/equipment/permit relationships.
- Quality & compliance audit agent cross-referencing inspection records against the regulatory corpus.

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